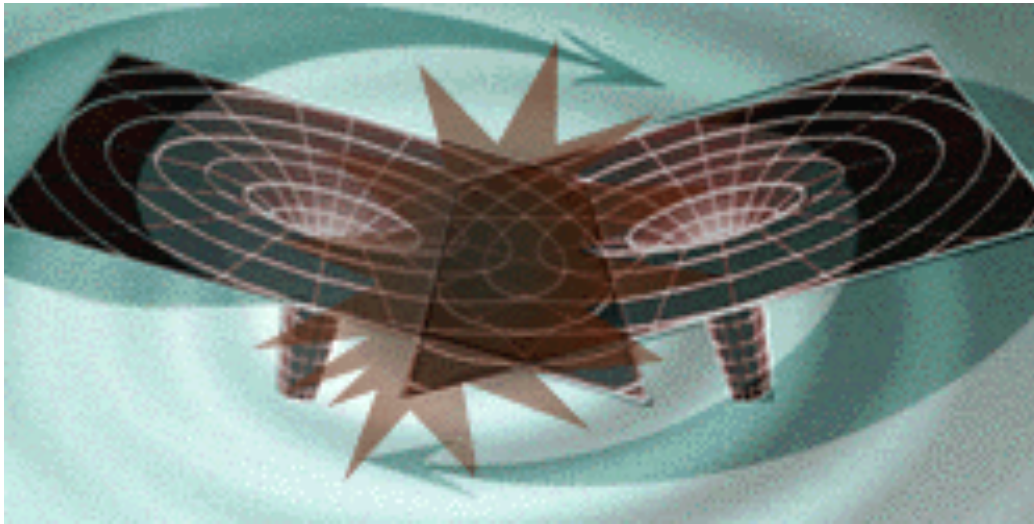


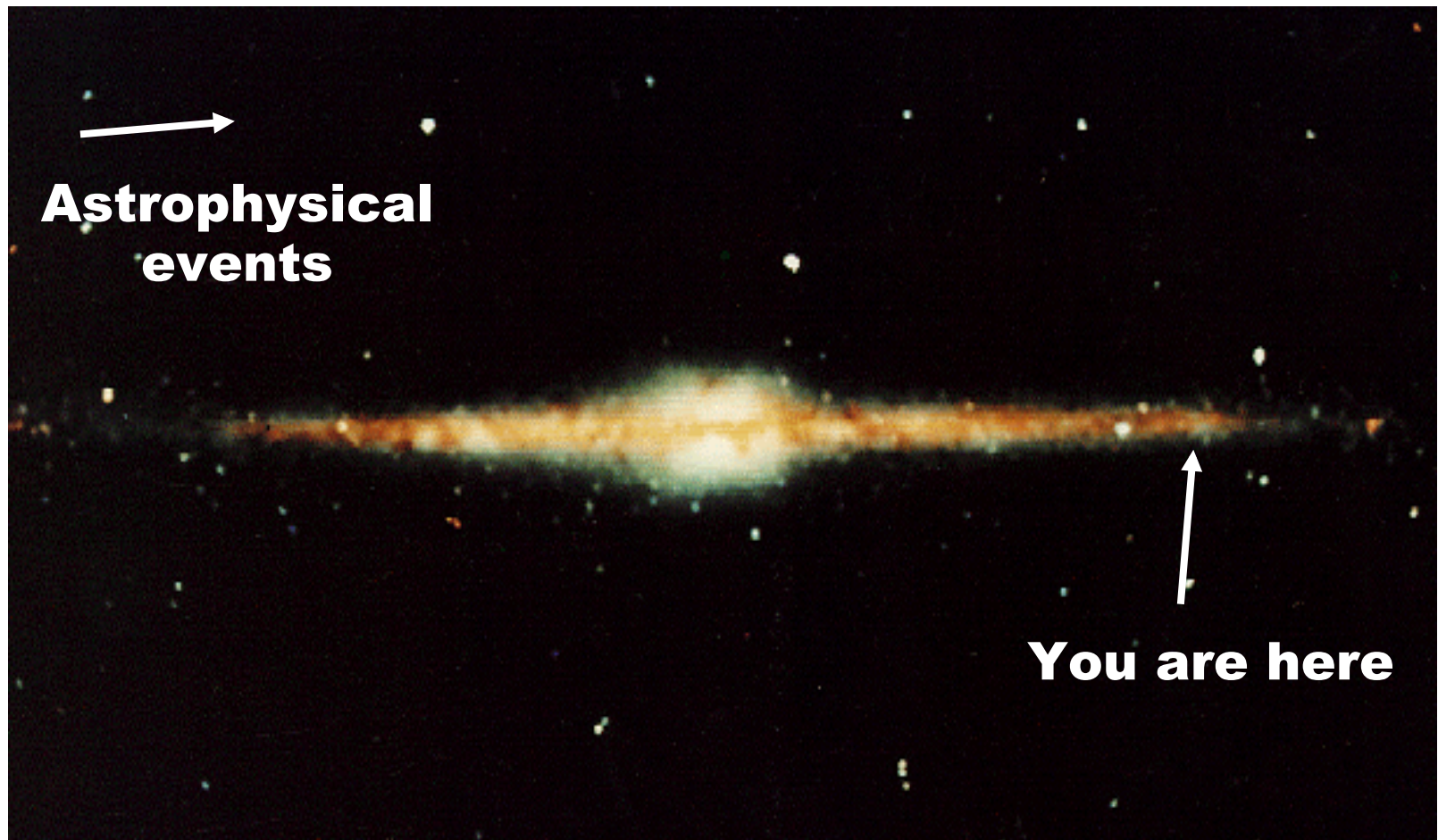
Computational (Gravitational Wave) Science using Sony PlayStations



***Gaurav Khanna
Associate Professor,
UMass Dartmouth, Physics***



The central problem of Astronomy:



The challenge ..

Conventional astronomy uses **electromagnetic radiation as its medium for observation. This poses several issues:**

- **Light can be easily shielded, distorted and dispersed.**
- **It is not really produced “by the astronomical bodies themselves”, but by the molecules on the surfaces of bodies. It can therefore produce a distorted view of what one is observing, due to a variety of reasons.**

The proposal: use gravitational waves.

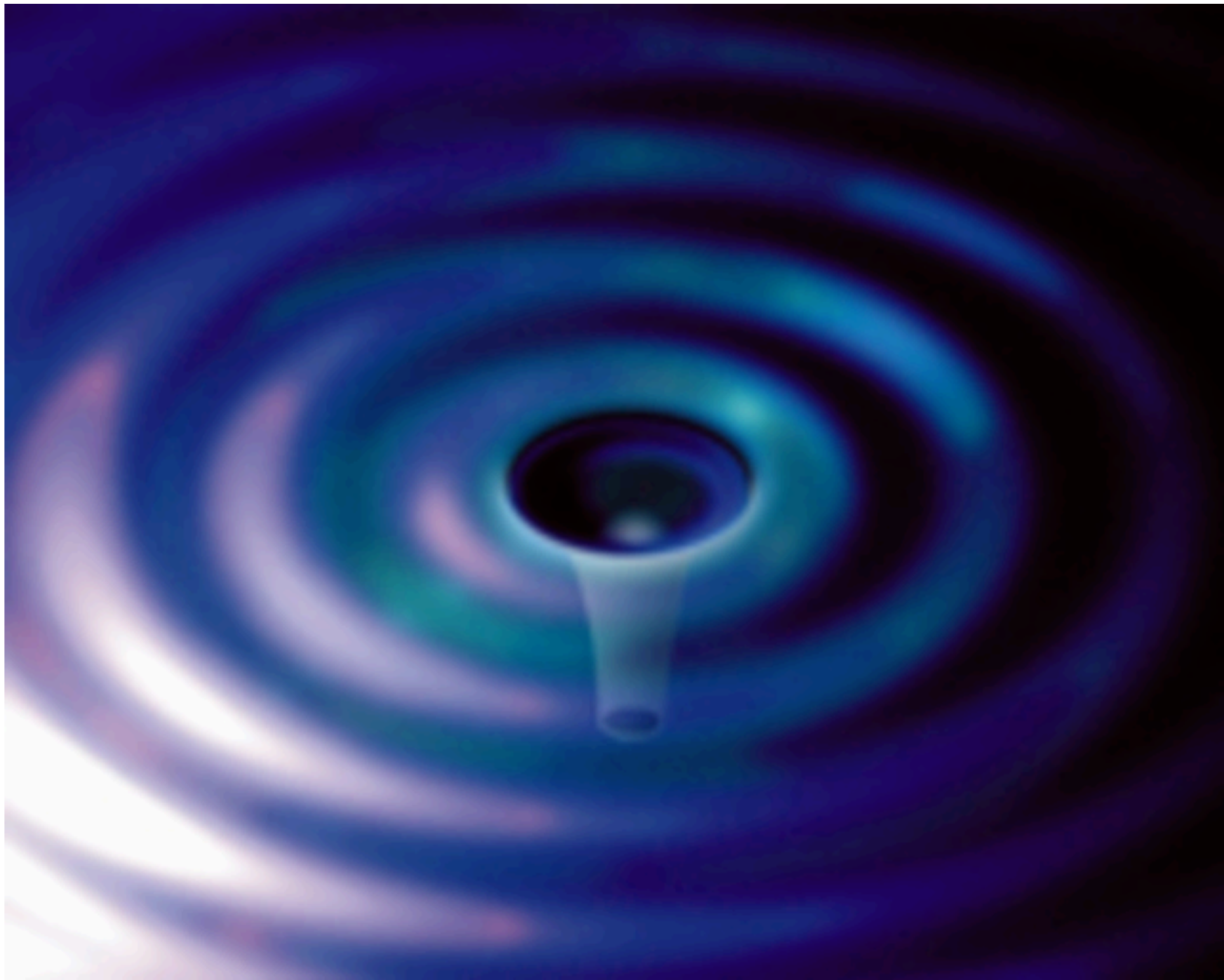
Gravitational waves

Are disturbances in the gravitational field that propagate at the speed of light. They are produced by accelerating masses.

They carry energy, momentum & angular momentum.

They are analogous to electromagnetic waves.

Just like light is a “disturbance” in the electromagnetic field, gravitational waves are “disturbances” or “ripples” in space-time itself.



Why gravitational waves?

Gravitational waves are produced by bulk matter, correlating much better with the “relevant” aspects of astronomical systems.

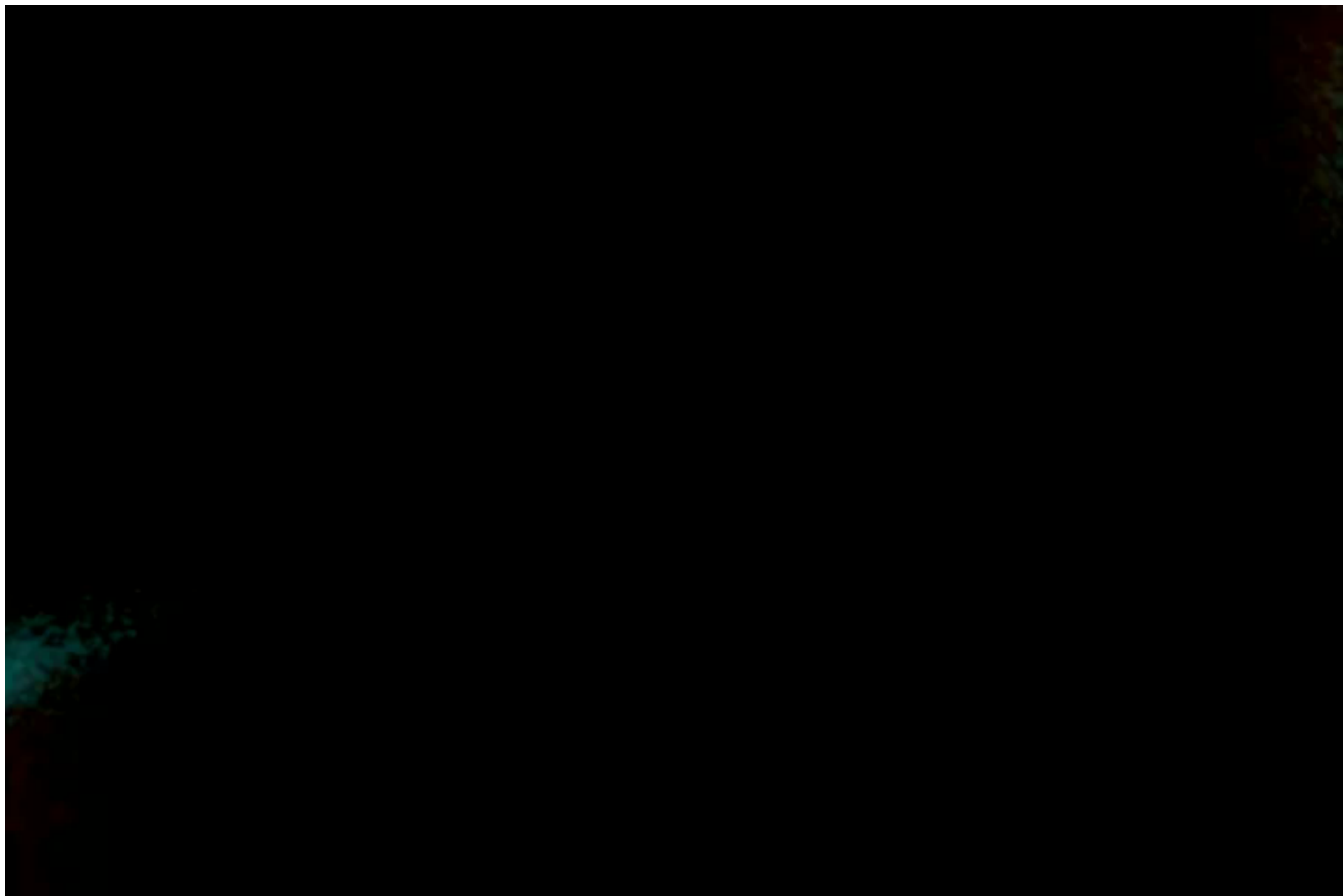
They are almost impossible to diffuse or disperse (unless large bulks of matter get in the way).

Sources of GWs

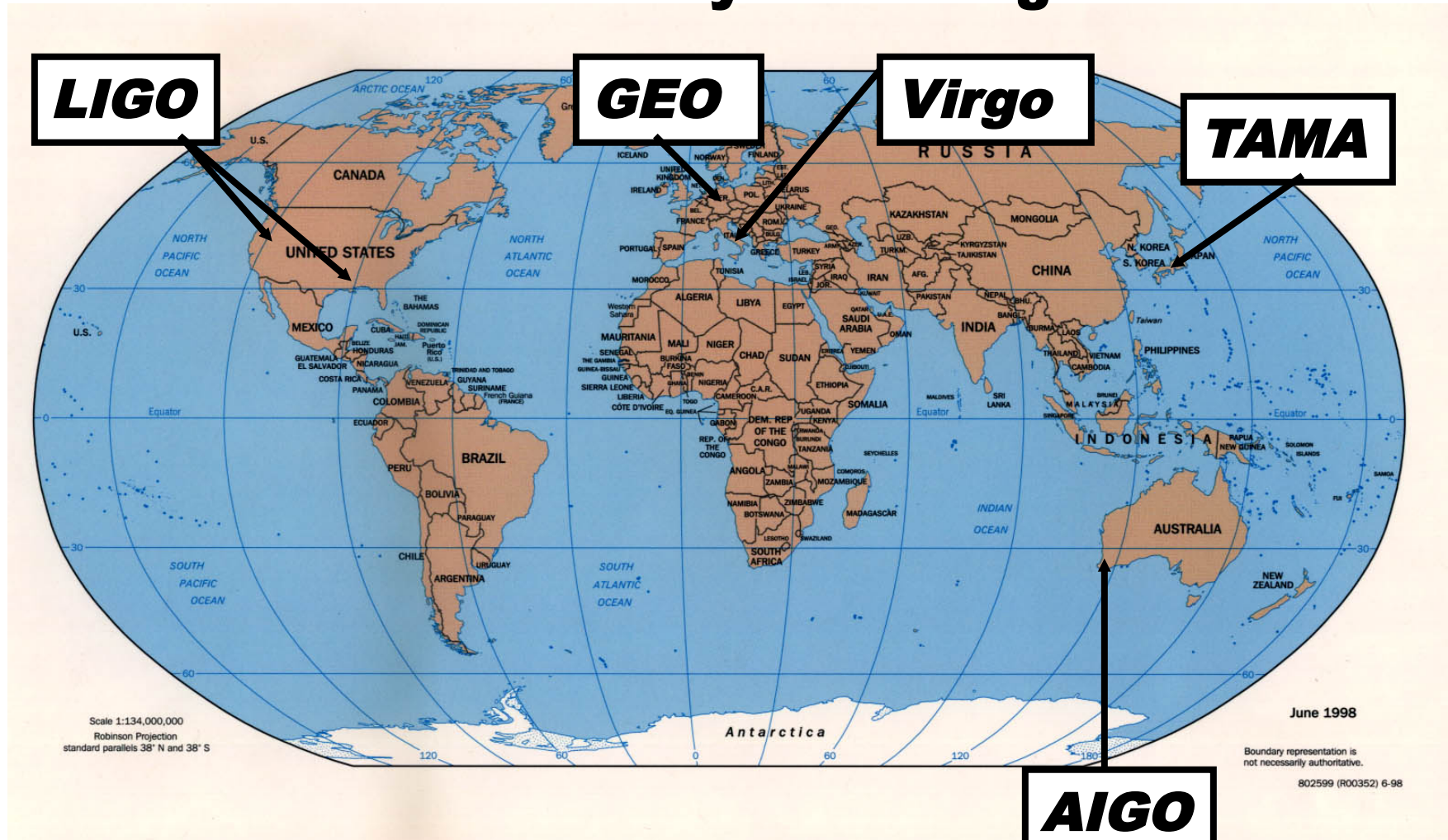
The most promising sources of gravitational waves are those that move dense objects at high accelerations. Examples are **supernovae and collisions of compact objects** like neutron stars and black holes.

Even for such exotic and violent events, the effects on Earth would be very weak. For instance, the very final burst of gravitational waves produced in **a collision of two solar mass black holes at 100 Mpc would cause a deformation or “strain” on Earth of less than 10^{-21} .**

This means, that one needs to be able to measure fluctuations of about 10^{-21} meters over a distance of a meter ..



Detection on Earth simultaneously detect signal



Gravitational wave detection

free masses

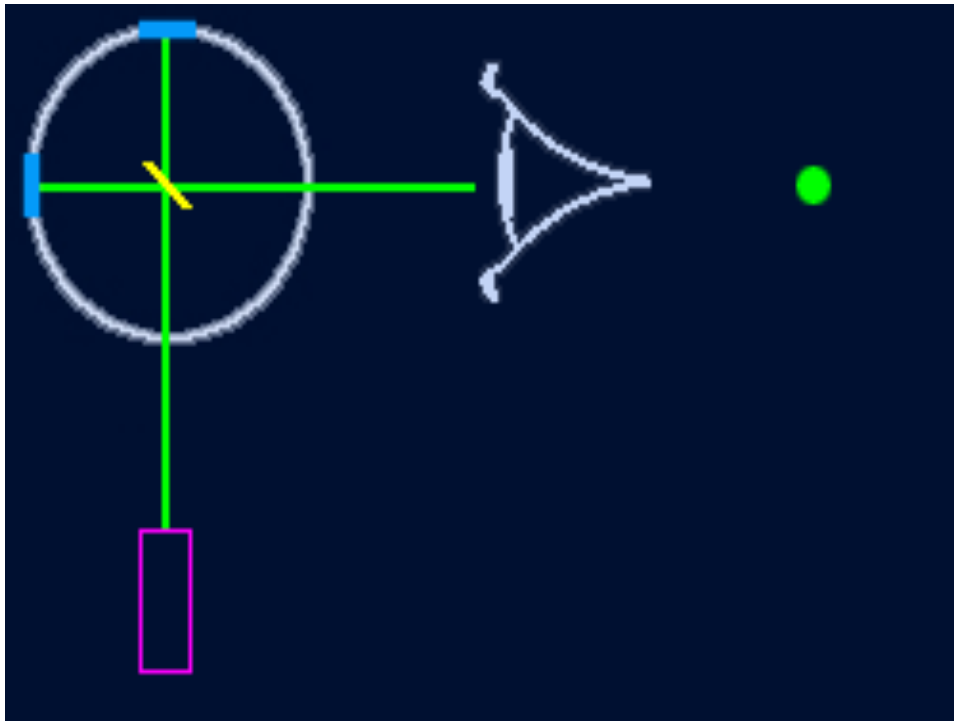


h = strain amplitude of grav. waves

$$h = \Delta L / L \sim 10^{-21}$$

$$L = 4 \text{ km}$$

$$\Delta L \sim 10^{-18} \text{ m}$$



**Laser
Interferometer**

Laser Interferometer Gravitational-wave Observatory (LIGO)

**Hanford
Observatory**

Caltech

MIT

**Livingston
Observatory**



LIGO

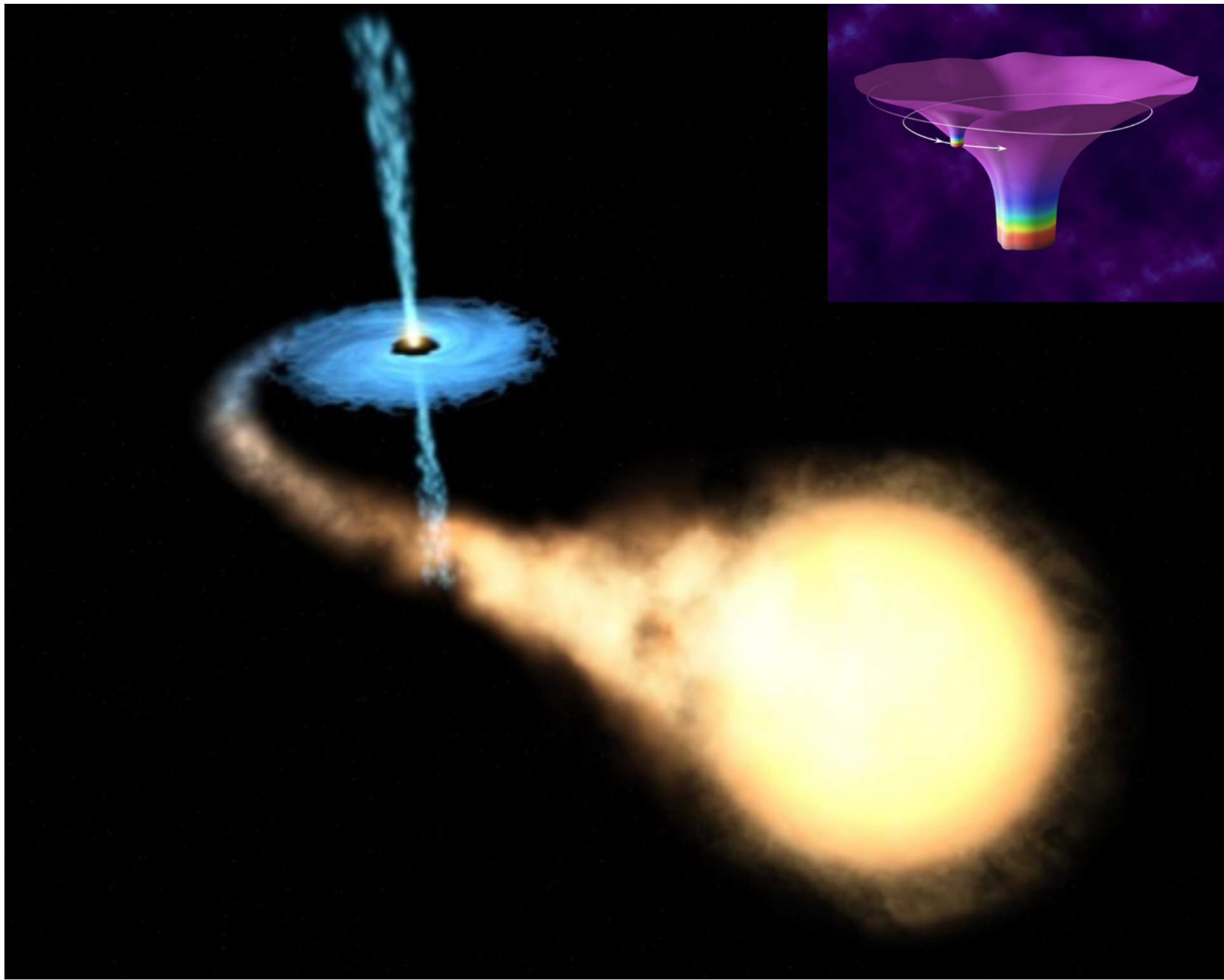
**Livingston,
Louisiana**



4 km

GWs from Black Holes

- **EMRI: extreme-mass-ratio inspiral**
- **Galactic nuclei have supermassive black holes (a million solar masses or so)**
- **What happens when a small star is captured by such a hole?**
- **Can we see the gravitation wave that would be “burped” out of the black hole?**
- **The ESA/NASA LISA Mission will be able to detect this radiation!**



Einstein's Equation

$$\begin{aligned}
 & \left[\frac{(r^2 + a^2)^2}{\Delta} - a^2 \sin^2 \theta \right] \frac{\partial^2 \psi}{\partial t^2} + \frac{4Mar}{\Delta} \frac{\partial^2 \psi}{\partial t \partial \phi} \\
 & + \left[\frac{a^2}{\Delta} - \frac{1}{\sin^2 \theta} \right] \frac{\partial^2 \psi}{\partial \phi^2} - \Delta^{-s} \frac{\partial}{\partial r} \left(\Delta^{s+1} \frac{\partial \psi}{\partial r} \right) \\
 & - \frac{1}{\sin \theta} \frac{\partial}{\partial \theta} \left(\sin \theta \frac{\partial \psi}{\partial \theta} \right) - 2s \left[\frac{a(r - M)}{\Delta} + \frac{i \cos \theta}{\sin^2 \theta} \right] \frac{\partial \psi}{\partial \varphi} \\
 & - 2s \left[\frac{M(a^2 - r^2)}{\Delta} - r - i a \cos \theta \right] \frac{\partial \psi}{\partial t} \\
 & + [s^2 \cot^2 \theta - s] \psi = 4\pi(r^2 + a^2 \cos^2 \theta)T
 \end{aligned} \tag{1}$$

where $\Delta = r^2 - 2Mr + a^2$ and $T = 2\rho^{-4} T_4$, where

$$\begin{aligned}
 T_4 = & (\Delta + 3\gamma - \gamma^* + 4\mu + \mu^*)[(\delta^* - 2\tau^* + 2\alpha)T_{nm^*} \\
 & - (\Delta + 2\gamma - 2\gamma^* + \mu^*)T_{m^*m^*}] + (\delta^* - \tau^* + \beta^* + 3\alpha \\
 & + 4\pi)[(\Delta + 2\gamma + 2\mu^*)T_{nm^*} \\
 & - (\delta^* - \tau^* + 2\beta^* + 2\alpha)T_{nn}].
 \end{aligned} \tag{2}$$

“source-term” T4

$$\begin{aligned}
 T := & 1/2/(r^2+a^2*\cos(\theta)^2)*((r^2+a^2)*(1/2*(r+\text{Complex}(1)*a*\cos(\theta)))/(r^2+a^2*\cos(\theta)^2)^{1/2}*(\text{Complex}(-1/16)*nmu*(r1(t)+\text{Complex}(1)*a*\cos(\theta1(t))))/(r1(t)^2+a^2*\cos(\theta1(t))^2)^{1/2}/(E*((a^2+r1(t)^2)^2/(r1(t)^2+a^2-2*M*r1(t))-a^2*\sin(\theta1(t))^2)+a*Iz*(1-(a^2+r1(t)^2)/(r1(t)^2+a^2-2*M*r1(t))))*(E*(a^2+r1(t)^2)+diff(r1(t),t)*(E*((a^2+r1(t)^2)^2/(r1(t)^2+a^2-2*M*r1(t))-a^2*\sin(\theta1(t))^2)+a*Iz*(1-(a^2+r1(t)^2)/(r1(t)^2+a^2-2*M*r1(t))))-a*Iz*(\text{Complex}(1)*\sin(\theta1(t))*(a*E-Iz/\sin(\theta1(t))^2)+diff(\theta1(t),t)*(E*((a^2+r1(t)^2)^2/(r1(t)^2+a^2-2*M*r1(t))-a^2*\sin(\theta1(t))^2)+a*Iz*(1-(a^2+r1(t)^2)/(r1(t)^2+a^2-2*M*r1(t))))))/Pi^2/wr*\exp(-1/2*(r-r1(t))^2/wr^2)/wt^3*(\theta-\theta1(t))*\exp(-1/2*(\theta-\theta1(t))^2/wt^2)*mm*diff(\phi1(t),t)*\exp(\text{Complex}(1)*mm*(\phi-\phi1(t)))/\sin(\theta1(t))+1/16*nmu*(diff(r1(t),t)+\text{Complex}(-1)*a*\sin(\theta1(t))*diff(\theta1(t),t))/(r1(t)^2+a^2*\cos(\theta1(t))^2)^{1/2}/(E*((a^2+r1(t)^2)^2/(r1(t)^2+a^2-2*M*r1(t))-a^2*\sin(\theta1(t))^2)+a*Iz*(1-(a^2+r1(t)^2)/(r1(t)^2+a^2-2*M*r1(t))))*(E*(a^2+r1(t)^2)+diff(r1(t),t)*(E*((a^2+r1(t)^2)^2/(r1(t)^2+a^2-2*M*r1(t))-a^2*\sin(\theta1(t))^2)+a*Iz*(1-(a^2+r1(t)^2)/(r1(t)^2+a^2-2*M*r1(t))))-a*Iz*(\text{Complex}(1)*\sin(\theta1(t))*(a*E-Iz/\sin(\theta1(t))^2)+diff(\theta1(t),t)*(E*((a^2+r1(t)^2)^2/(r1(t)^2+a^2-2*M*r1(t))-a^2*\sin(\theta1(t))^2)+a*Iz*(1-(a^2+r1(t)^2)/(r1(t)^2+a^2-2*M*r1(t))))))/Pi^2/wr*\exp(-1/2*(r-r1(t))^2/wr^2)/wt^3*(\theta-\theta1(t))*\exp(-1/2*(\theta-\theta1(t))^2/wt^2)*\exp(\text{Complex}(1)*mm*(\phi-\phi1(t)))/\sin(\theta1(t))-1/8*nmu*(r1(t)+\text{Complex}(1)*a*\cos(\theta1(t)))/(r1(t)^2+a^2*\cos(\theta1(t))^2)^{3/2}/(E*((a^2+r1(t)^2)^2/(r1(t)^2+a^2-2*M*r1(t))-a^2*\sin(\theta1(t))^2)+a*Iz*(1-(a^2+r1(t)^2)/(r1(t)^2+a^2-2*M
 \end{aligned}$$

$$\begin{aligned}
& * \text{diff}(\text{th1}(t), t) + \text{diff}(\text{diff}(\text{th1}(t), t), t) * (E * ((a^2 + r1(t)^2)^2 / (r1(t)^2 + a^2 - 2 * M * r1(t)) - a^2 * \sin(\text{th1}(t))^2) + a * lz * (1 - (a^2 + r1(t)^2) / (r1(t)^2 + a^2 - 2 * M * r1(t)))) + \text{diff}(\text{th1}(t), t) * (E * (4 * (a^2 + r1(t)^2) / (r1(t)^2 + a^2 - 2 * M * r1(t)) * r1(t) * \text{diff}(r1(t), t) - (a^2 + r1(t)^2)^2 / (r1(t)^2 + a^2 - 2 * M * r1(t))^2 * (2 * r1(t) * \text{diff}(r1(t), t) - 2 * M * \text{diff}(r1(t), t)) - 2 * a^2 * \cos(\text{th1}(t)) * \sin(\text{th1}(t)) * \text{diff}(\text{th1}(t), t)) + a * lz * (-2 * r1(t) * \text{diff}(r1(t), t) / (r1(t)^2 + a^2 - 2 * M * r1(t)) + (a^2 + r1(t)^2) / (r1(t)^2 + a^2 - 2 * M * r1(t))^2 * (2 * r1(t) * \text{diff}(r1(t), t) - 2 * M * \text{diff}(r1(t), t))))) / \text{Pi}^2 / wr * \exp(-1/2 * (r - r1(t))^2 / wr^2) / wt^3 * (\text{theta} - \text{th1}(t)) * \exp(-1/2 * (\text{theta} - \text{th1}(t))^2 / wt^2) * \exp(\text{Complex}(1) * mm * (\text{phi} - \text{phi1}(t))) * \sin(\text{th1}(t)) + 1/16 * nm * (r1(t) + \text{Complex}(1) * a * \cos(\text{th1}(t))) / (r1(t)^2 + a^2 * \cos(\text{th1}(t))^2)^{1/2} / (E * ((a^2 + r1(t)^2)^2 / (r1(t)^2 + a^2 - 2 * M * r1(t)) - a^2 * \sin(\text{th1}(t))^2) + a * lz * (1 - (a^2 + r1(t)^2) / (r1(t)^2 + a^2 - 2 * M * r1(t)))) * (E * (a^2 + r1(t)^2) + \text{diff}(r1(t), t) * ((a^2 + r1(t)^2)^2 / (r1(t)^2 + a^2 - 2 * M * r1(t)) - a^2 * \sin(\text{th1}(t))^2) + a * lz * (1 - (a^2 + r1(t)^2) / (r1(t)^2 + a^2 - 2 * M * r1(t)))) - a * lz * (\text{Complex}(1) * \sin(\text{th1}(t)) * (a * E - lz / \sin(\text{th1}(t))) + \text{diff}(\text{th1}(t), t) * (E * ((a^2 + r1(t)^2)^2 / (r1(t)^2 + a^2 - 2 * M * r1(t)) - a^2 * \sin(\text{th1}(t))^2) + a * lz * (1 - (a^2 + r1(t)^2) / (r1(t)^2 + a^2 - 2 * M * r1(t)))))) / \text{Pi}^2 / wr^3 * (r - r1(t)) * \text{diff}(r1(t), t) * \exp(-1/2 * (r - r1(t))^2 / wr^2) / wt^3 * (\text{theta} - \text{th1}(t)) * \exp(-1/2 * (\text{theta} - \text{th1}(t))^2 / wt^2) * \exp(\text{Complex}(1) * mm * (\text{phi} - \text{phi1}(t))) / \sin(\text{th1}(t)) - 1/16 * nm * (r1(t) + \text{Complex}(1) * a * \cos(\text{th1}(t))) / (r1(t)^2 + a^2 * \cos(\text{th1}(t))^2)^{1/2} / (E * ((a^2 + r1(t)^2)^2 / (r1(t)^2 + a^2 - 2 * M * r1(t)) - a^2 * \sin(\text{th1}(t))^2) + a * lz * (1 - (a^2 + r1(t)^2) / (r1(t)^2 + a^2 - 2 * M * r1(t))))
\end{aligned}$$

$$\begin{aligned}
& 2)^2/(r_1(t)^2+a^2-2*M*r_1(t))-a^2*\sin(\theta_1(t))^2+a*l_z*(1-(a^2+r_1(t)^2)/(r_1(t)^2+a^2-2*M*r_1(t))))/Pi^2/wr*\exp(-1/2*(r-r_1(t))^2/wr^2)/wt*\exp(-1/2*(\theta-\theta_1(t))^2/wt^2)*\exp(\text{Complex}(1)*m*(\phi-\phi_1(t)))/\sin(\theta_1(t))*(2*r_1(t)*\text{diff}(r_1(t),t)-2*a^2*\cos(\theta_1(t))*\sin(\theta_1(t))*\text{diff}(\theta_1(t),t))*(\text{Complex}(1)*\cos(\theta_1(t))*\text{diff}(\theta_1(t),t)*(a*E-l_z/\sin(\theta_1(t))^2)+\text{Complex}(2)/\sin(\theta_1(t))^2*l_z*\cos(\theta_1(t))*\text{diff}(\theta_1(t),t)+\text{diff}(\text{diff}(\theta_1(t),t),t)*(E*((a^2+r_1(t)^2)^2/(r_1(t)^2+a^2-2*M*r_1(t))-a^2*\sin(\theta_1(t))^2)+a*l_z*(1-(a^2+r_1(t)^2)/(r_1(t)^2+a^2-2*M*r_1(t))))+\text{diff}(\theta_1(t),t)*(E*(4*(a^2+r_1(t)^2)/(r_1(t)^2+a^2-2*M*r_1(t))*r_1(t)*\text{diff}(r_1(t),t)-(a^2+r_1(t)^2)^2/(r_1(t)^2+a^2-2*M*r_1(t))^2*(2*r_1(t)*\text{diff}(r_1(t),t)-2*M*\text{diff}(r_1(t),t))-2*a^2*\cos(\theta_1(t))*\sin(\theta_1(t))*\text{diff}(\theta_1(t),t))+a*l_z*(-2*r_1(t)*\text{diff}(r_1(t),t)/(r_1(t)^2+a^2-2*M*r_1(t))+(a^2+r_1(t)^2)/(r_1(t)^2+a^2-2*M*r_1(t))^2*(2*r_1(t)*\text{diff}(r_1(t),t)-2*M*\text{diff}(r_1(t),t))))))-1/4*nmu*(r_1(t)+\text{Complex}(1)*a*\cos(\theta_1(t))^2/(r_1(t)^2-a^2*\cos(\theta_1(t))^2)^3/(E*((a^2+r_1(t)^2)^2/(r_1(t)^2+a^2-2*M*r_1(t))-a^2*\sin(\theta_1(t)^2)+a*l_z*(1-(a^2+r_1(t)^2)/(r_1(t)^2+a^2-2*M*r_1(t))))*(\text{Complex}(1)*\sin(\theta_1(t)*E-l_z/\sin(\theta_1(t))^2)+\text{diff}(\theta_1(t),t)*(E*((a^2+r_1(t)^2)^2/(r_1(t)^2+a^2-2*M*r_1(t))-a^2*\sin(\theta_1(t))^2)+a*l_z*(1-(a^2+r_1(t)^2)/(r_1(t)^2+a^2-2*M*r_1(t))))^2/Pi^2)*\exp(-1/2*(r-r_1(t))^2/wr^2)/wt*\exp(-1/2*(\theta-\theta_1(t))^2/wt^2)*\exp(\text{Complex}(1)*m*(\phi-\phi_1(t)))/\sin(\theta_1(t))*(2*\text{diff}(r_1(t),t)^2+2*r_1(t)*\text{diff}(\text{diff}(r_1(t),t),t)+2*a^2*\sin(\theta_1(t))^2*\text{diff}(\theta_1(t),t)^2-2*a^2*\cos(\theta_1(t))^2*\text{diff}(\theta_1(t),t)^2-2*
\end{aligned}$$

... and lots more!

- Our simulations would require several 100 computer processors**
- Supercomputers are expensive!**
- Recently, we had a neat idea ..**

PlayStation 3 (!!)



“PS3 Gravity Grid”



How powerful?

Our 16 PS3 cluster is comparable to **over 100 processor-cores of a typical Linux compute cluster that uses Intel Xeon processors.**

Costs **order-of-magnitude less!**
(\$5K / \$50K)

IT WORKS GREAT FOR US!

Why the PS3?

- **Open platform (runs Linux!)**
- **Powerful Cell processor**
“Supercomputer-on-a-chip”
- **Very low cost (below cost price!)**
- **Readily available in large quantities (mass produced!)**

Media attention ..

- **ABC News**
- **Wired Magazine**
- **New Scientist Mag**
- **MSNBC**
- **Space.com**
- **Washington Post**
- **NPR**
- **Science Channel**



Some Results

